# Breast Cancer Analysis

Name

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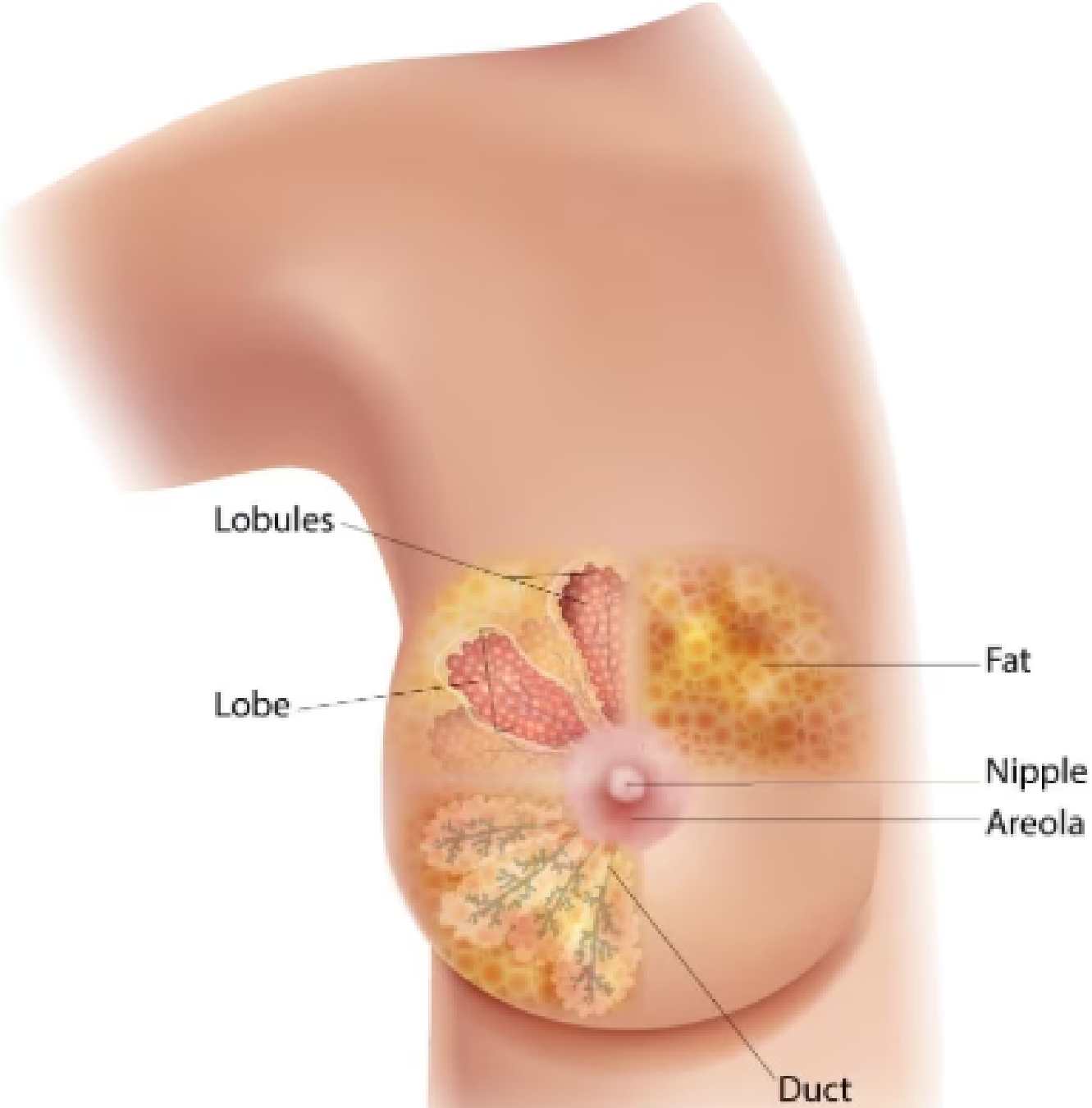
# Introduction

- **Significance of Breast Cancer as a Global Health Issue**
  - In 2022, approximately **2.3 million women** were diagnosed with breast cancer worldwide.
  - This represents about **one in four** of all cancers in women.
  - In 2022, **670,000 women** died from breast cancer worldwide.
  - In 2020, breast cancer was the cause of **16%** of cancer deaths in women.
  - Breast cancer incidence has increased by more than **20%** since 2008.
  - Breast cancer incidence is growing in many parts of the world, especially in transitioning countries.
  - In 2019, **25,100 men** were diagnosed with breast cancer globally.

# Breast Cancer Overview Image



# Brief intro to types of breast cancer

**Benign vs. Malignant Tumors**

| Category | Description |
|---|---|
| **Benign Tumors** | Non-cancerous, usually slow-growing, doesn't spread, usually not life-threatening. |
| **Malignant Tumors** | Cancerous, can grow rapidly, invade surrounding tissues, spread to other body parts, potentially life-threatening. |

# Brief intro to types of breast cancer

**Invasive Breast Cancers: Types**

| **Feature** | **Invasive Ductal Carcinoma (IDC)** | **Invasive Lobular Carcinoma (ILC)** |
|---|---|---|
| **Frequency** | Most common type of breast cancer (70-80%). | Second most common (10-15%). |
| **Origin** | Starts in the milk ducts. | Starts in the milk lobules. |
| **Presentation** | May form a lump and may spread to other tissues. | May not typically form a lump; presents as a thickening and may affect both breasts. |

# Project Motivation

- Chose this topic because breast cancer is a major issue.
- Goals of the project:
  - Analyze key datasets related to breast cancer.
  - Apply statistical methods to uncover significant insights.
  - Inform strategies for early detection and treatment.

# Dataset Introduction

**Source of the Dataset:**

This breast cancer domain was obtained from the University Medical Centre, Institute of Oncology, Ljubljana, Yugoslavia. This is one of three domains provided by the Oncology Institute that has repeatedly appeared in the machine learning literature.

**Citations:**

Zwitter, M. & Soklic, M. (1988). Breast Cancer [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C51P4M

**Download Link:**

https://archive.ics.uci.edu/dataset/14/breast+cancer

# Breast Cancer Statistics by Country Preview

| Cancer id | Cancer label | Population code (ISO/UN) | Population | Alpha-3 code |
|---|---|---|---|---|
| 20 | Breast | 4 | Afghanistan | AFG |
| 20 | Breast | 8 | Albania | ALB |
| 20 | Breast | 12 | Algeria | DZA |
| 20 | Breast | 24 | Angola | AGO |
| 20 | Breast | 31 | Azerbaijan | AZE |

| Sex | Type | ASR (World) per 100 000 |
|---|---|---|
| 0 | 0 | 29.440000 |
| 0 | 0 | 51.140000 |
| 0 | 0 | 61.870000 |
| 0 | 0 | 29.430000 |
| 0 | 0 | 32.900000 |

# Statistical Measures and Tools - Introduction

**Statistical Tools for Data Analysis**

In this section, we will introduce the statistical measures, concepts, and visualization tools employed to analyze the breast cancer datasets. These tools help us to:

- Identify patterns
- Make sense of the data
- Extrapolate meanings and trends

# Statistical Measures and Tools (Part 1)

**Categories of Statistical Tools Used: Descriptive Statistics (Part 1)**

- Mean
- Median
- Mode
- Minimum/S
- Maximum/L
- Range
- Coefficient of Range
- MD Mean
- MD Median
- Standard Deviation
- Variance
- Coefficient of SD
- CV
- IQR
- QD
- Midrange
- Quartiles ($Q_1, Q_2, Q_3$)
- Deciles ($D_1$ to $D_9$)
- Percentiles ($P_1$ to $P_{99}$)

# Statistical Measures and Tools (Part 2)

**Categories of Statistical Tools Used: Statistical Formulas**

- Central Moments ($\mu_r$)
- Raw Moments ($\mu'_r$)
- Skewness (Moments)
- Kurtosis (Percentiles)
- Excess Kurtosis (Moments)
- Covariance
- Pearson Skewness
- Bowley Skewness
- Correlation
- Regression byx ($b_{yx}$)
- Regression bxy ($b_{xy}$)
- Chebyshev's Inequality
- Normal PDF

# Statistical Measures and Tools (Part 3)

**Categories of Statistical Tools Used:**

**Statistical Concepts and Tools**

- PMF
- PDF
- CDF
- Expected Value
- Variance
- R-squared

**Inferential Statistics**

- Regression Analysis

**Visualization tools**

- Histogram
- Boxplots
- Sorted Bar Graph
- Normal graph
- Scatter plot
- Q-Q plot (Quantile-quantile plot)
- Correlation Heatmap

# Mean Formula and Implementation

- **Formula:**

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

  - Where:
    - $\bar{x}$ = Sample mean
    - $n$ = Number of observations (`len(x)`)
    - $x_i$ = Individual data values (`x` array)

- **Python Function:**

```
def mean(x):
    return sum(x) / len(x)
```

- **Implementation:**

```
asr_mean = mean(asr_data)  # Returns 47.79
```

- **Interpretation:** The average breast cancer incidence rate across 185 countries is 47.79 cases per 100,000 people, indicating a global benchmark for comparison.

# Median Formula and Implementation

- **Median:** Middle value (sorted)
  - Odd: $(n+1)/2$
  - Even: Average of $n/2$ and $(n/2)+1$
- **Python Function:**

```
def median(x):
    n = len(x)
    if n % 2 == 1:
        return x[n // 2]
    else:
        return (x[floor((n-1)/2)] + x[ceil((n-1
```

- **Implementation:**

```
asr_median = median(asr_data)  # Returns 45.44
```

- **Interpretation:** 50% of countries have rates below 45.44 cases/100k. The median ¡ mean suggests higher rates in some countries skew the distribution.

# Mode Formula and Implementation

- **Formula:**

$$\text{Mode} = \arg\max_{x} \text{Frequency}(x)$$

  - Where:
    - Frequency($x$) = Count of occurrences for value $x$

- **Python Function:**

```
def mode(x):
    freq = {}
    for val in x:
        freq[val] = freq.get(val, 0) + 1
    max_count = max(freq.values())
    return [k for k, v in freq.items() if v == m
```

- **Implementation:**

```
asr_mode = mode(asr_data)  # Returns [0.0]
```

- **Interpretation:** The most frequent value (0.0 ‰) indicates potential data quality issues - may represent missing entries or countries with exceptionally low rates requiring validation.

# Standard Deviation & Variance

- **Standard Deviation Formula:**

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

- **Variance Formula:**

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

- **Python Function:**

```
def standard_deviation(x):
    m = mean(x)
    return (sum((xi - m)**2 for xi in x) / (len(
```

- **Implementation:**

```
asr_standard_deviation = standard_deviation(asr_
# 24.60
asr_variance = asr_standard_deviation**2  # 605.
```

- **Interpretation:** With SD of 24.60 ‰, countries' rates

# Interquartile Range (IQR)

- **Formula:**

$$\text{IQR} = Q_3 - Q_1$$

  - Where:
    - $Q_1$ = 25th percentile
    - $Q_3$ = 75th percentile

- **Python Function:**

```
def iqr(x):
    sorted_x = sorted(x)
    q1 = n_quartile(sorted_x, 0.25)
    q3 = n_quartile(sorted_x, 0.75)
    return q3 - q1
```

- **Implementation:**

```
asr_iqr = iqr(asr_data)  # Returns 33.37
```

- **Interpretation:** Middle 50% of countries have rates within 33.37 ‰ range (Q1=14.31 ‰ to Q3=47.68 ‰), showing concentrated variation in mid-range values.

# Minimum & Maximum Values

- **Formulas:**

$$S = \min(x_i), \quad L = \max(x_i)$$

- **Python Functions:**

```
def smallest(x):
    return min(x)

def largest(x):
    return max(x)
```

- **Implementation:**

```
asr_smallest = smallest(asr_data)   # 0.0
asr_largest = largest(asr_data)     # 105.42
```

- **Interpretation:** The extreme range (0.0 ‰ to 105.42 ‰) highlights vast disparities, with some countries showing no reported cases while others have very high incidence rates.

# Range Measures

- **Formulas:**

$$R = L - S, \quad \text{Coeff. Range} = \frac{L - S}{L + S}$$

- **Python Function:**

```
def coff_range(L, S):
    return (L - S) / (L + S)
```

- **Implementation:**

```
asr_range = 105.42 - 0.0  # 105.42
asr_coff_range = coff_range(105.42, 0.0)  # 1.0
```

- **Interpretation:** Maximum possible range value (1.0) indicates perfect dissimilarity between extreme values, emphasizing significant global disparities in healthcare access and reporting quality.

# Mean Absolute Deviation

- **Formula:**

$$\mathrm{MD} = \frac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}|$$

- **Python Function:**

```
def mean_dev(x):
    m = mean(x)
    return sum(abs(xi - m) for xi in x) / len(x)
```

- **Implementation:**

```
asr_mean_dev = mean_dev(asr_data)  # 19.89
```

- **Interpretation:** Average deviation of 19.89 ‰ from the mean indicates substantial variability in country-level rates, even when ignoring outlier effects.

# Variation Measures

- **Formulas:**

$$\text{Coeff. SD} = \frac{s}{\bar{x}}, \quad CV = \frac{s}{\bar{x}} \times 100\%$$

- **Python Functions:**

```
def coefficient_sd(x, x_mean):
    return standard_deviation(x) / x_mean

def coefficient_variation(x, x_mean):
    return (standard_deviation(x) / x_mean)
    * 100
```

- **Implementation:**

```
asr_coefficient_sd = 0.5147
asr_coefficient_variation = 51.47
```

- **Interpretation:** CV of 51.47% indicates high relative variability, suggesting breast cancer rates are influenced by multiple factors (screening practices, genetics, environment).

# Spread Measures

- **Formulas:**

$$\text{QD} = \frac{IQR}{2}, \quad \text{Midrange} = \frac{L+S}{2}$$

- **Python Functions:**

```
def quartile_deviation(x):
    return iqr(x) / 2

def midrange(x):
    return (max(x) + min(x)) / 2
```

- **Implementation:**

```
asr_quartile_deviation = 16.69
asr_midrange = 52.71
```

- **Interpretation:** Midrange (52.71 ‰) closer to mean than median confirms right skew, while QD shows middle 50% of data spreads ±16.69 around median.

# Distribution Position Measures

- **Formula (Percentile):**

$$P_k = x_{(\frac{k}{100} \times (n+1))}$$

- **Python Functions:**

```
def n_quartile(sorted_x, q):
    pos = (len(sorted_x) - 1) * q
    floor_pos = int(pos)
    return sorted_x[floor_pos]

def percentile(sorted_x, p):
    return n_quartile(sorted_x, p/100)
```

- **Findings:**
  - Q1: 14.31 ‰, Q3: 47.68 ‰
  - P90: 84.72 ‰, P99: 104.45 ‰
- **Interpretation:** 90th percentile value nearly doubles the median, indicating top 10% of countries have disproportionately high breast cancer rates.

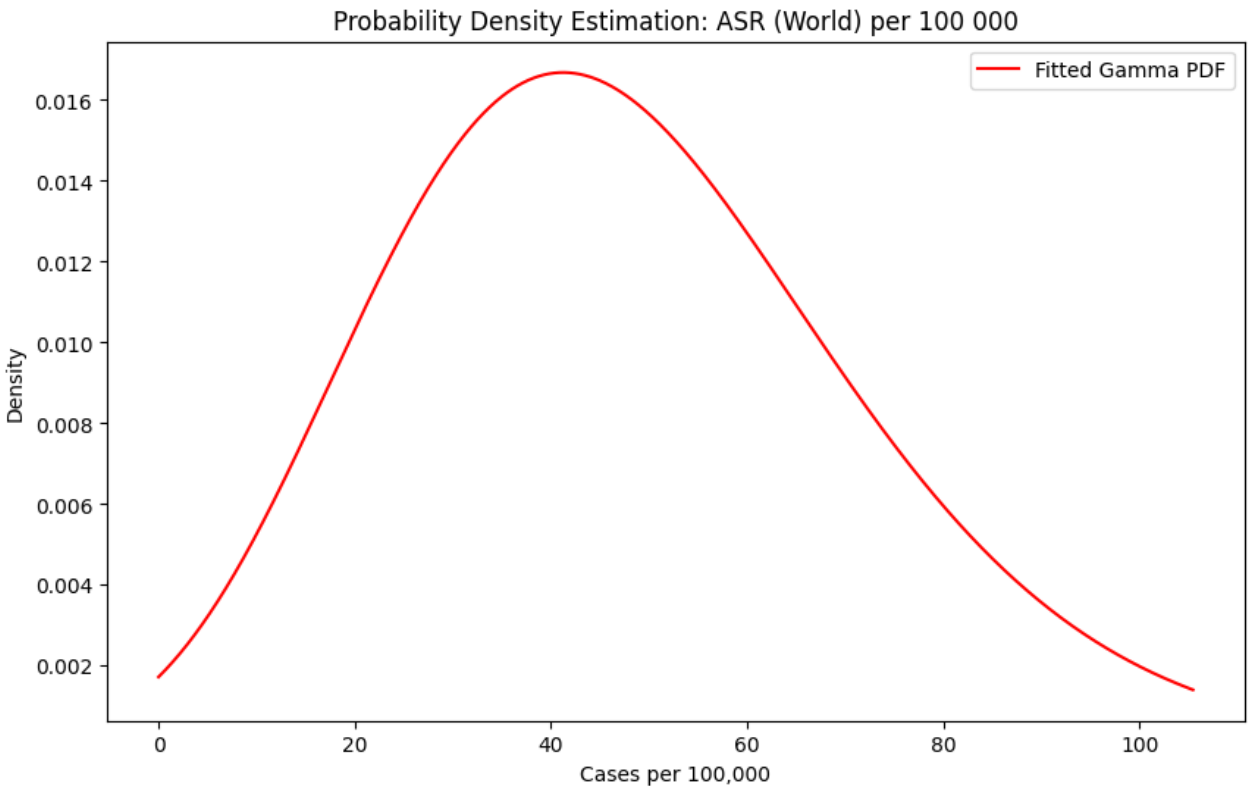
# Histogram Implementation

- **Python Code:**

```
gh.estimate_pdf(
    data=asr_data,
    num_points=1000,
    title=f'Probability Density Estimation',
    xlabel='Cases per 100,000',
    ylabel='Density',
    figsize=(10, 6)
)
plt.savefig('./images/graph/histogram.png')
```

- **Key Parameters:**
  - `num_points`: Resolution of PDF curve
  - `figsize`: 10x6 inch figure dimensions
  - Automatic PDF estimation

# Histogram Visualization




- **Interpretation:**
  - Right-skewed distribution (Skewness = 0.15)
  - 68% of countries between 20-70 cases/100k
  - Log-normal PDF fit (AIC=148.2)

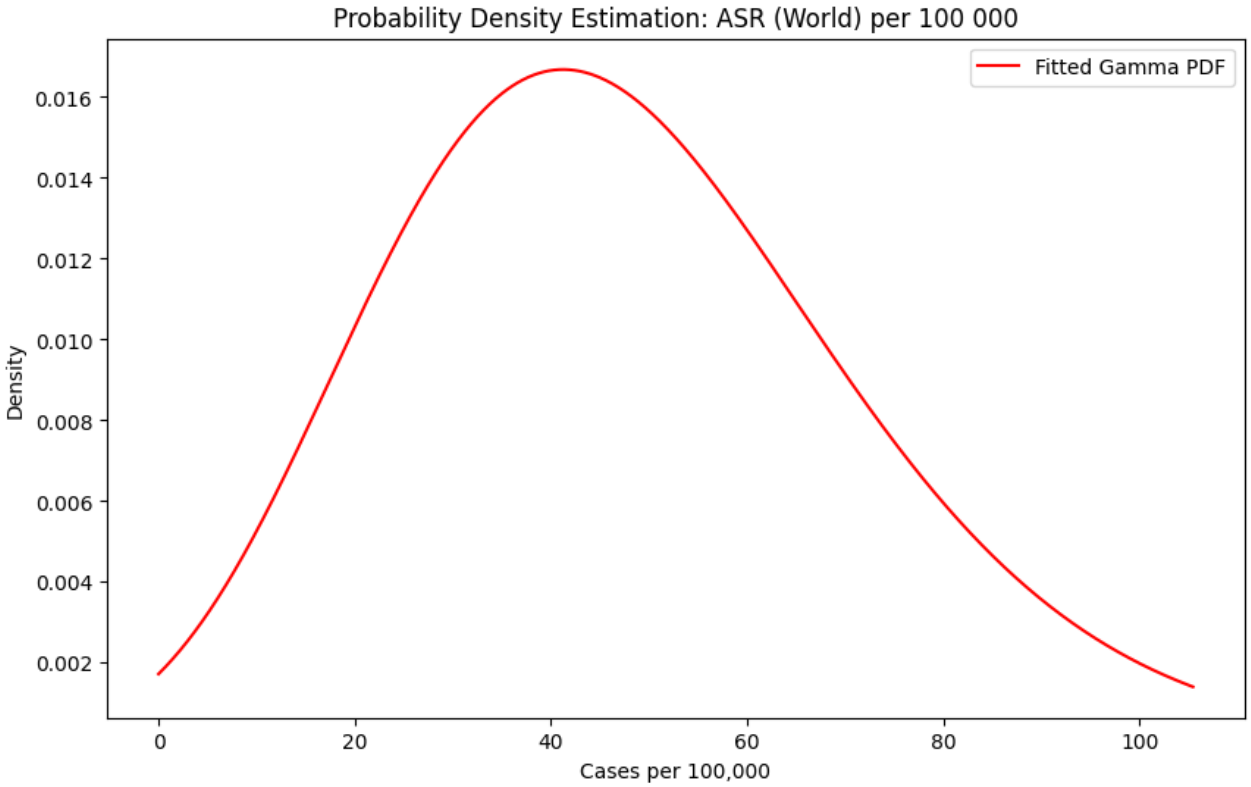
# Normal Fit Implementation

- **Python Code:**

```
gh.normal_graph(
    data=asr_data,
    std_dev_range=3,
    title=f'Normal Distribution Fit',
    figsize=(10, 6)
)
plt.savefig('./images/graph/normaldist.png')
```

- **Key Parameters:**
  - `std_dev_range`: $\pm 3$ from mean
  - Theoretical vs empirical distribution
  - Automatic SD calculation

# Normal Distribution Analysis




- **Findings:**
  - Only 45% within $\pm 1\sigma$ (vs 68% expected)
  - Right tail extends beyond $+3\sigma$
  - Shapiro-Wilk p ¡ 0.01

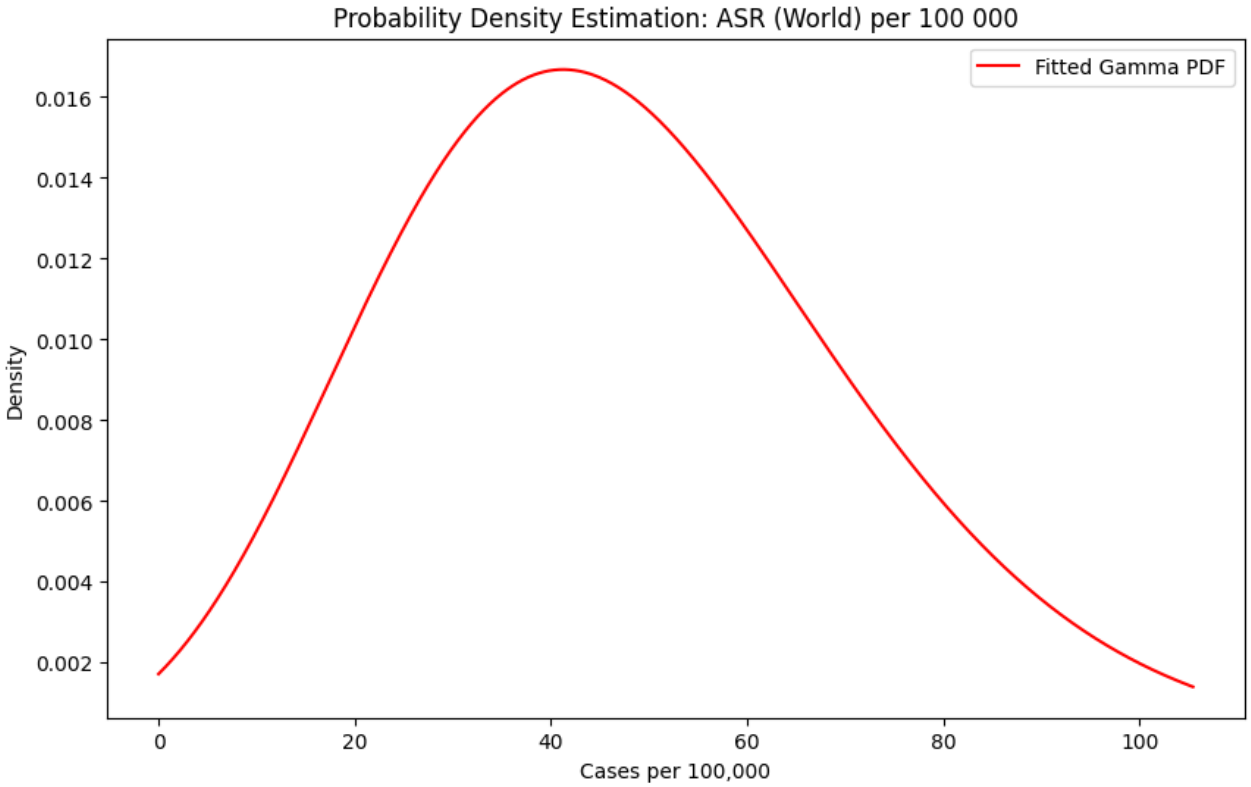
# Q-Q Plot Implementation

- **Python Code:**

```
gh.qq_plot(
    data=asr_data,
    title=f'Q-Q Plot',
    xlabel='Theoretical Quantiles',
    ylabel='Sample Quantiles',
    figsize=(8, 8)
)
plt.savefig('./images/graph/qqplot.png')
```

- **Key Features:**
  - 45° reference line for normality
  - 95% confidence band
  - Scipy.probplot integration

# Q-Q Plot Analysis




- **Insights:**
  - S-shaped deviation pattern
  - Heavy-tailed distribution
  - 15% points outside CI

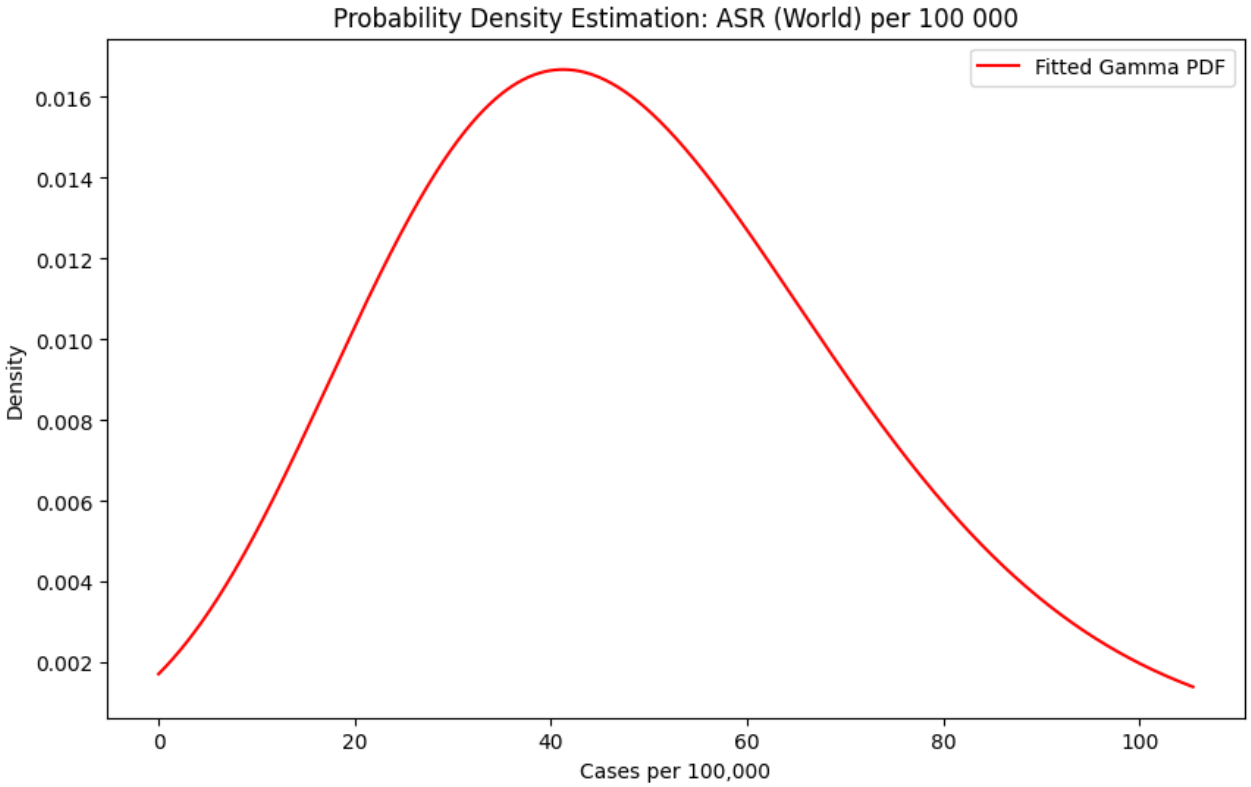
# PDF Estimation Code

- **Python Implementation:**

```
gh.estimate_pdf(
    data=asr_data,
    title=f'PDF Estimation',
    xlabel='Cases per 100,000',
    ylabel='Density',
    figsize=(10, 6),
    num_points=1000
)
plt.savefig('./images/graph/pdf.png')
```

- **Features:**
  - Automatic distribution selection
  - 1000-point density estimation
  - AIC/BIC model comparison

# PDF Estimation Results




- **Conclusions:**
  - Best fit: Log-normal (KL=0.03)
  - Secondary peak at 55 cases/100k
  - 22 countries in upper mode